Zenith: Learning a Financial Reasoning Pipeline from Daily Experience

The Lab, Oracle and Evolve system

Will Cook*

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Abstract

Zenith is a financial reasoning system built around repeated experience. A historical trading day's available feeds enter Lab, an inspectable reasoning graph that produces a cited forecast or an explicit no-call. After the forecast horizon, Oracle compares the frozen attempt with later observations and investigates which earlier evidence could have supported a better decision. Evolve uses that comparison to propose a transferable change to the reasoning pipeline; a reviewed successor then faces later days. The learning object is the pipeline's instructions, context, skills and eventually structure. The current implementation retains the original full Lab and Oracle, source-bound agent handoffs, output contracts and versioned policies in a standalone source checkout. A pinned three-equity panel provides 291 shared daily price dates and 4,990 date-bucketed news summaries, with explicit availability assumptions. The latest primary-agent development sequence completed two full Labs and Oracles. After a declared price-history rule lost to unchanged-price persistence, Evolve proposed reducing its scale; the next Lab used that stored change before its outcomes were inspected. The changed rule then beat persistence but lost to the unchanged rule's arithmetic control. Earlier attempts exposed unsupported causal reasoning, forced forecasting and forced criticism; the contracts now permit explicit abstention and a testable empirical hypothesis without an invented cause. An audit of a richer public source found future-year results in historical news rows. These findings establish an executed experience-to-procedure transition and specific failures, not a learned forecasting advantage or a measured benefit from news. The next empirical question is whether reviewed changes improve later reasoning and forecasts beyond an unchanged pipeline.

1 The system we want to learn

The design starts from the job a financial reasoner should repeatedly perform: inspect the available information, explain a view, make commitments, confront what happens and improve the procedure. Historical data supplies many dated instances of that job. It should preserve the shape of the intended live inputs rather than determine an unrelated research task simply because a benchmark is convenient to score.

Lab, Oracle and Evolve implement three distinct responsibilities. Lab commits using the information available at the issue time. Oracle receives both that frozen attempt and

*AI agents researched, implemented and drafted this work under Will Cook's direction. Recorded source material and execution evidence delimit its claims.

later evidence. Evolve decides whether the experience justifies a change that can be used without future information on a subsequent attempt. The investigator may be the lead agent or a worker given a fixed prompt and the same bound evidence. A scheduled chat is not required by the algorithm; it is one possible execution service.

Here, learning initially means changing durable reasoning procedures and context policies, not updating the underlying language model's weights. Useful changes might make a miner recover the last comparable disclosure, require a synthesis node to confront a contrary observation, or make a forecaster disclose how it converted a qualitative view into a target price. Graph changes are a broader development option. One short skill update is the first implemented daily transition, not a permanent limit on the programme.

The central distinction is between improving a historical answer and improving the procedure that produces later answers. A hindsight explanation may expose a missed earlier signal, but it may also depend entirely on information that arrived afterward. That difference must survive into the proposed lesson.

The operator's generation analogy comes from AlphaGo Zero: experience generated by a current system informs an improved policy, which supports subsequent experience [23]. Here the policy is a reasoning procedure, and its candidate version is compared with the incumbent on the same later market situations. This is a design analogy, not an implementation of neural self-play or tree search. Market outcomes must come from external observations; replaying a historical date cannot create a new independent market realization.

2 One native graph, three stages

The standalone checkout contains the existing loader, executor, authored node JSON, contracts and policy machinery. It does not depend on the parent workspace's kernel, Task Ledger, private dossiers or interactive Station. The physical registry has 71 nodes across all retained mission families. The daily path executes the full 28-node Lab and 31-node Oracle; it does not substitute the earlier three-node price-only experiment.

The original engine assembles prompts, validates dependency order and accepts reasoning responses through its native importer. A missing response creates a pending packet; it is not successful reasoning. The daily controller advances only after the preceding mission is complete. Lab output remains fixed while Oracle examines it. Rejected responses, failed provider attempts and uncertain cost reservations remain part of the record.

The temporal handoff between versions is not a cycle inside a single forward graph. The first Lab cannot acquire Oracle's later inputs through an ordinary edge. Oracle keeps the original issue anchor and the later target time; Evolve's successor is used only on the next eligible issue. Historical replay still does not simulate the wall-clock time required to obtain and review real closing data. A live system needs a corresponding execution delay.

Stage	Evidence and work	Persistent result
Snapshot	Completed prices and eligible news at one trading-day boundary	Immutable feeds, clocks, source hashes and missingness
Lab	Two mining passes, cross-lane synthesis, CP1 thesis, CP2 forecasts and integrity review	Frozen predictions, supporting trace and validation results
Oracle	Original attempt, original evidence and separate later observations	Reconciliation, hindsight brief, critique and ideal CP2
Evolve	Bound Lab/Oracle pair and current policy	Reviewed no-change decision or versioned candidate
Later Lab	A later snapshot and selected policy	New forecasts whose behavior can be assessed

Table 1: The daily loop. Completion of a stage and quality of its reasoning are separate facts.

2.1 A prediction can be tested before its cause is known

The original causal thesis contract prevented a forecast when a dominant market force was unresolved. That is appropriate for rejecting a fabricated causal explanation, but it also prevented testing a separately declared empirical mapping. The explicit `EMPIRICAL` branch now permits that test alongside the existing causal and abstention modes.

The first implemented operator is deliberately a price-only control. For each declared stock, it averages the last five daily simple returns, multiplies the mean by a frozen scale and clips the result at a declared magnitude cap. Multiplying the last observed price by one plus that return gives the next-close target. The full graph still processes the available news; that text does not silently enter this formula. Thus the control tests execution and parameter transfer, not the incremental value of news or complex reasoning.

CP1 declares the input citations, target set, history length, scale and cap; the engine recomputes the result from the original source history and checks consecutive XNYS dates, issuer identity and the next-session horizon. The Director cannot alter that rule after CP1. Oracle also retains its parameters and numeric forecast rather than replacing them with a hindsight optimum. The status remains explicitly unvalidated. Richer empirical operators require their own declared input mapping; this initial rule is not a general financial model or a restriction on future development.

3 Historical feeds that fit the actual system

The longer development adapter uses three JSON files from InvestorBench [15, 16]. The pinned MSFT, HON and JNJ files share all 291 XNYS sessions from 12 March 2020 through 6 May 2021. They contain 3,636, 444 and 910 news summaries respectively. Our source audit finds no missing shared price sessions; all three symbols have news on 196 dates. These counts describe the selected files, not reproduced benchmark performance.

A snapshot contains twenty observed scalar prices per symbol and the preceding seven calendar days of complete upstream summaries. Summary date buckets lack publication and capture clocks, so the adapter assumes availability at midnight in New York on the following calendar day and excludes the current date. It excludes all filing summaries because their availability is unestablished. This is a reconstructed development stream: the lag does not prove historical availability, issuer relevance, adjustment vintage or immunity to model memory. The source repository's MIT license also does not independently establish rights to redistribute every publisher-derived item. Public code supplies the pins and adapter; downloaded text remains in local storage.

The initial adapter uses the original AA price and article files in FNSPID, pinned to source revision `4054842` [10]. For each selected XNYS session it emits the last 20 completed daily price bars and full eligible article bodies from the preceding seven calendar days. The source is a practical starting slice: two small files, approximately 8.66 MB together. It does not supply a complete multi-asset research universe.

STOCK and NEWS contain observations. ETF, MACRO, POLYMARKET, CALCULATOR and STOCKGRID retain their native artifact names but explicitly contain no supplied observations. Their absence limits support; it is not an economic event. A market article discussing macro conditions does not become an independently observed macro series. Preserving these distinctions permits the full graph to run while making the cost of sparse coverage visible.

Source admission uses original publication strings rather than blindly trusting upstream preprocessing. For example, a January article marked 17:51 EST is 22:51 UTC and must not enter that day's 21:00 UTC close snapshot. The inspected upstream converter shifted that offset in the wrong direction. Literal `Text=0` is missing text, not neutral news. Invalid or conflicting clocks are excluded and counted. The adapter retains original body hashes and stable evidence identifiers; an AA filename does not establish that every article concerns Alcoa specifically.

The Jan16,17,18 2024 snapshots contain 1,7,7 eligible article bodies, respectively. Jan18 has no fresh article in this slice; repeated context is not new evidence. Source prices use `Close` consistently. Corporate-action vintages, original capture times and later article revisions are unverified. Filtering a publisher clock alone cannot establish complete point-in-time integrity. Modern language models may also remember historical events [11].

Other data candidates remain available for investigation. An inspected FinMultiTime article slice had body/date/URL mismatches and was not admitted to this replay; that observation does not invalidate separately sourced price or filing tables [26]. Data acquisition remains explicit. The public code does not redistribute these article bodies.

A richer candidate, the public Herculean database, contains 2,656 price rows, 3,888 news summaries and 73 filing excerpts; its eight issuers share 332 XNYS dates [13]. A fixed inspection of the earliest three MSFT dates found two temporal counterexamples. News row 2432, dated 2 December 2024, reports $77.7 billion revenue and $3.72 EPS matching Microsoft's October 2025 release. Row 2433, dated 3 December 2024, explicitly describes FY2026 Q2 and $81.3 billion revenue announced in January 2026 [18, 19]. A one-day lag cannot repair future-year content. These rows justify quarantining this news

source, not an estimate of its global contamination rate or rejection of every price and filing record. The pinned file hash, queries and row identities are in `docs/herculean-audit.json`.

4 Oracle: outcome, explanation and earlier opportunity

Oracle first asks a mechanical question: what happened to each declared target over its declared horizon? Exact later observations score the frozen forecast. It then asks interpretive questions: which original claims were supported, which information was missed, what appeared only afterward, and which proposed causes remain uncertain? A correct direction does not validate the explanation that accompanied it.

An ideal historical answer is useful only when its earlier citations are actually available at the earlier boundary. Later reporting may direct the investigation toward a neglected earlier signal, but it cannot itself become a Lab citation. A surprising event with no earlier support should remain an unforecastable or uncertain component rather than an instruction that earlier miners ought to have discovered the future. Outcome-only and richer hindsight teachers are distinct experimental conditions, related to learning with privileged information [24].

A useful teacher must also be able to report no evidenced error. The native contracts now distinguish a cited Lab abstention, an unexplained realized move and a review that finds no concrete earlier mistake. CP1 locks the forward decision mode; an abstained CP2 has no predictions. Oracle retains observed price changes without creating prediction errors or a hit rate. Its explicit no-findings review must name the alternatives examined and the limits of its evidence. None of these outcomes proves optimality. They permit Evolve to retain a procedure instead of accumulating invented lessons.

Financial interpretation must also distinguish familiar facts from changes. For example, Salesforce's May 2023 release preserved annual revenue guidance while raising non-GAAP operating-margin guidance [21, 22]. That supports a profitability revision, not a revenue revision or an assertion about analyst surprise. Adobe's June release raised some annual targets while lowering a segment revenue range [1–3]. These supporting cases test a recurring reasoning skill; they are not the selected replacement for the daily system.

The remaining price effect is a further question. A public announcement may already be reflected in the price at which a decision could be executed. Headline-return work motivates separating observed response from later predictability [17]. Economic relationships can motivate indirect transmission [8]; shared words alone do not establish one.

5 Evolve: reviewed transitions between versions

The existing Evolve runner exports its fixed analysis instructions, original Lab CP2 and Oracle quartet with source bindings. The daily controller adds a small response contract: diagnosis, uncertainty, proposed later test, exact source quotations, and either one Lab skill change or an explicit no-change decision. The lead agent can perform this review directly. A worker receives the same packet and returns the same shape.

Import checks the packet identity, reconstructs source bindings, resolves each quoted JSON field against its original artifact, validates the permitted node and compiles a native reasoning policy. It retains rejected attempts. An accepted candidate is a new policy file; the next day's Lab freezes that file in its execution manifest. The original policy and dossiers remain available. Repeated continuation reads recheck the decision and candidate instead of trusting a success label alone.

These checks establish provenance, structure and an explicit review decision. They cannot establish that a quoted fact supports the proposed generalization. The reviewing agent's identity is a disclosure, not authenticated expertise. No-change is a legitimate outcome when the episode supplies no convincing lesson. The transition is development adoption for later testing, not promotion to a proven live trading policy.

TextGrad and GEPA provide relevant precedents for textual feedback and reflective prompt search [4, 31]; Trace and multi-stage instruction optimization motivate inspecting intermediate computations [7, 20]. Financial agent systems provide precedents for layered memory and persistent guidance [29, 30]. Zenith uses these sources to sharpen its own explicit transition and controls. It does not claim that implementing a loop establishes a new optimization method or inherits their reported results.

6 What has actually executed

The latest sequence used the three-equity InvestorBench snapshots at the 24 and 27 July 2020 closes, with outcomes at the 27 and 28 July closes. Both Labs completed all 28 nodes and both Oracles all 31, using 64 accepted primary-agent reasoning responses in total. No prior-run responses were replayed and no paid provider calls were made. One Oracle response used prose in an enum field; it was rejected, retained and corrected. This was a disclosed historical development sequence, not an independent evaluation of an automated provider.

The initial rule used five returns, scale 0.25, cap 0.02 and all three stocks. After its error exceeded persistence, Evolve proposed halving only the scale. The reviewed addon appears in the next Lab's actual Decide prompt, and its frozen numerical forecasts use scale 0.125. The agent then inspected the next outcomes. Table 2 reports the resulting errors.

Issue → outcome	Native scale	Native	Unchanged	Persistence
24 → 27 July	0.25	0.98649	0.98649	0.93047
27 → 28 July	0.125	0.62531	0.59575	0.65488

Table 2: Mean absolute error in percentage points of return: $100|\widehat{P} - P_{\text{later}}|/P_{\text{issue}}$, averaged across the three stocks. Unchanged scale 0.25 and persistence are deterministic arithmetic controls on the same inputs, not separately executed native arms.

The first issue had one correct direction; the second had two. The changed rule beat persistence on the second issue but lost to the unchanged rule. Shrinkage helped the

wrong-direction HON forecast and reduced useful magnitude for both correct directions. These are three correlated assets on each of only two dates. The second Evolve review made no further change and selected no winner. Its no-change receipt preserves experimental lineage, not a promotion of the current candidate. No generation-retention command was run. The source-bound receipt, `docs/empirical-pilot-receipt.json`, includes the forecasts, outcomes, reviews and successor prompt binding.

This is an actual feedback-to-procedure transition with a negative comparison. It establishes neither useful news inference nor an advantage of the full graph over the transparent formula. Executing a larger cohort of the same formula through many reasoning nodes would not by itself address those gaps.

6.1 Earlier development evidence

A paid full-graph development attempt using DeepSeek V4.1 Flash stopped at CP1 after 17 of 28 Lab nodes and 11 reasoning calls. Its responses confused artifact names with valid evidence identifiers, misstated numerical facts and turned missing lanes into market narrative. The attempt and its accounting remain preserved; it is not counted as a completed prediction.

A separate primary-agent attempt completed all 28 Lab nodes through 21 native reasoning imports. Its paired Oracle completed all 31 nodes through 11 accepted reasoning imports. The original forecast was down from approximately $27.87 to $27.50; the later close was approximately $27.18, producing absolute price error of $0.32. Direction was correct. Semantic review remained red because an observed trend did not establish a forward causal mechanism. This is one exposed, manually reasoned development episode, not a performance estimate for the failed paid model or the wider system.

The primary review proposed that the director distinguish the measured pattern, proposed mechanism and numerical target rule, and avoid increasing causal confidence after one directional hit. A following-day Lab was prepared with that policy but has no completed successor forecast. The newer daily controller makes the same kind of review and transition repeatable. Native hydration and synthetic transition tests establish its wiring; they do not supply additional historical forecast results. The original episode is recorded in `docs/daily-replay-receipt.json`.

A second development case used the three-equity InvestorBench snapshot at the 1 July 2020 close and later prices from 2 July. The earlier packet contained 73 summaries; inspection separated reported operating facts from summary sentiment, duplicate reports and indirect company mentions. It did not establish a supported next-session directional thesis. The old contracts nevertheless required a dominant cause and forecast, and Oracle later required a nonempty critique. These were engineering defects exposed by the pilot, not learned market discoveries.

The explicit abstention, unresolved-cause and no-findings branches permit the full native loop to represent this case honestly. The final execution completed 28 Lab and 31 Oracle nodes with 21 and 11 accepted response imports. Twenty-nine answers were exact reuses from the preceding development attempts; three Oracle responses were newly authored under the repaired contracts. The record is a contract replay, not 32 independent model samples or an untouched assessment. MSFT, HON and JNJ scalar prices

rose approximately 0.7621, 0.6315 and 0.4203 percent. No prediction loss or hit rate exists because Lab abstained. Oracle retained the no-call and Evolve recorded a reviewed no-change decision. The source-bound receipt is `docs/abstention-receipt.json`. Neither this review nor the contract repair establishes a financial edge.

Earlier A-share numerical and text controls remain useful negative development evidence [5, 12]. On 33 exposed dates, raw mean squared errors were 0.00141850 and 0.00148344 versus a zero-relative-return reference of 0.00109518. Chronological scaling produced 0.00108729 and 0.00108654. The design followed earlier aggregate inspection; the small differences do not establish an LLM or incremental text advantage. Their retained diagnosis concerns forecast magnitude, not completion of the full daily learning loop.

7 The next useful experiment

The immediate objective is competent repeated execution with enough genuine information for the reasoning job. The newly admitted longer panel now supports this work, alongside review of missing and stale evidence and a model that consistently meets the existing contracts. Increasing the number of agents does not repair a weak source packet or an unsupported target rule.

The implemented generation adapter freezes a reviewed challenger and the exact policy that produced its teacher episode, using the existing native cohort executor and feedback evaluator. Both run the full Lab on identical later snapshots and target support with the same disclosed model identity. The selection loss is mean issue-level mean target-price relative error. Its improvement margin is declared before execution. Complete support and at least twenty issue dates are required; these are decision gates, not significance guarantees. An abstention is a retained terminal attempt with missing forecast coverage, never a zero-loss success; this gate therefore cannot select an abstaining strategy. A selective strategy would require a separately declared coverage and utility objective. A twenty-date two-version comparison requires 840 reasoning responses before retries, which makes base competence a practical prerequisite.

A separately reviewed retention receipt binds the comparison, selected policy and final truth cutoff. The challenger is retained only if it clears the declared margin; otherwise the incumbent remains. The next generation's first issue must follow that cutoff. Historical development remains exposed, even when these chronology and provenance checks pass. No such generation comparison has yet completed; synthetic retention tests establish the control boundary only.

Once the base works, compare an unchanged pipeline with a pipeline that receives reviewed updates, on identical later snapshots and target support. Preserve all attempts and separate source failure, invalid output, abstention, forecast error and reasoning quality. Include simple price references and a strong direct reasoner before attributing benefit to graph complexity. Numerical calibration, experience retrieval and learned procedures need separate controls. Repeated selection on the same history creates optimism [6, 9, 27]; later untouched and prospective episodes are stronger evidence than another explanation of an inspected result.

The supplied temporal forecasting paper motivates combining time-series context and text [28]; explanatory prose still needs a fidelity test. Cross-stock semantic work suggests candidate relationships to verify [14]. Financial-network forecasting motivates numerical and narrative analogs, but its targets differ from stock returns [25]. Their ideas can improve source selection and diagnostic questions without turning their benchmarks into Zenith's objective.

A trading claim would additionally require an executable entry rule, delays, positions, overlapping exposures, distributions and costs. Neither the one-day manual replay nor the supporting A-share labels establish profitability. The current deliverable is a self-contained implementation of the original learning loop and an auditable place to improve it. Whether its changes produce better later financial reasoning remains the research question.

Reproduction. Start with `README.md` and `docs/daily-replay.md`. The source checkout supports editable installation on macOS/Linux; wheel and Windows distribution are not verified. `zenith daily init` freezes the daily inputs; `zenith daily next` exposes the next actual native task; `zenith import-response` validates Lab/Oracle answers; `zenith daily review` records the reviewed Evolve decision. `zenith daily matchoff` freezes a native comparison and `zenith daily retain` records its reviewed selection; the next initializer consumes `--retention`. See `docs/generations.md`. No command in this controller purchases inference or starts a separate chat. Datasets are reacquired by pinned source identity. Historical execution custody remains local where source rights prevent bundling full inputs and transcripts. `paper/README.md` maps the evidence and limitations.

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